

## IMPACT OF SCREEN TIME:

Vivaan Verma

GitHub Repository Link: <https://github.com/dg-vverma19/screen-time-impact-analysis->

### Summary of Research Questions and Results:

#### 1.) Does productivity level differ by screen time bracket?

- According to my data visualization, I discovered that the relationship between productivity level and screen time bracket is non-linear. Specifically, I first thought that there would be a very clear pattern in which lower screen time would mean more productivity. However, according to my stacked bar graph, I notice that the majority of the data is spread evenly across all screen time brackets. Overall, I discovered that that productivity level does not directly differ my screen time brackets through my dataset, but rather there might be multiple factors unaccounted for.

#### 2.) Does attention span level differ by screen time bracket?

- According to my data visualization, I discovered a relationship between attention span and screen time brackets. Specifically, as screen time increased, lower attention span counts in participants also increased. However, this was not a linear pattern, suggesting a more complex relationship that I will uncover in my analysis.

#### 3.) How does screen time differ by education levels?

- According to my data visualization, I discovered a relationship between attention span and education levels. Specifically, graduate participants increased as screen time increased, and undergraduate slightly decreased. This subtle relationship shows that students in higher education levels might link to daily screen time usage. However, this pattern is also not extremely clear, which will be further analyzed in my analysis.

#### 4.) Are certain age groups more likely to have a higher screen time?

- By creating a machine learning model, I discovered that higher age groups are more likely to have a higher screen time. I first believed that lower age groups would have higher screen time due to activities such as scrolling, or studying, but from this I uncovered participants who were 45 and above, actually dominated in each screen time interval.

### Motivation:

I care about this specific issue, as it is an event that has rapidly increased in significance recently. Specifically, as society continues to technologically develop, and develop access to more online tools, screen time inevitably increases, whether it's for work, studying, or personal entertainment. However, screen time is also a major issue in the younger generation of this era, as I have personally noticed this. These research questions uncover where screen time is most common statistically, and its clear impact in multiple aspects. Knowing the answers to these affects our understanding of this development, by observing the possible positive and negative consequences, and predicting what future generations might be doing cognitively. Analyzing this can also facilitate research on if future devices can be modified to incorporate more positive features, rather than negative.

## Dataset:

The dataset that I have chosen is from Kaggle and can be accessed through this link:

<https://www.kaggle.com/datasets/muhammalirazazaidi/screen-time-data-productivity-and-attention-span?resource=download>

Note: To download the dataset as a CSV file, one will have to sign into Kaggle first.

This dataset contains 15 descriptive columns including: age group, gender, education level, occupation, average daily screen time, device, screen activity, app category, screen time period, preferred work environment, productivity, attention span, work strategy, notification handling, and usage of products. For my research questions, I will not be utilizing all of these columns.

Overall, this dataset contains much data providing insight into the preferences and impact based on screen time, as well as the details about each respondent. This will overall allow for a deep analysis that will be effectively able to answer my research questions, as well as fulfill my motivation.

## Method:

RQ1: Does productivity level differ by screen time bracket?

- 1.) Group the dataset by screen time bracket and productivity level, then use groupby to count how many respondents fall into each combination.
- 2.) Using the Plotly library, create a single stacked vertical bar chart, with screen time bracket on the x-axis, respondent counts on the y-axis, and bars colored by productivity level.
- 3.) Compare the stacked segments across brackets to observe how the distribution of productivity levels shifts as screen time increases.

This will allow for a conclusion on the first research question, by visually analyzing how productivity is distributed as screen time increases, segment by segment within each bar. This also uses the challenge goal by using an external library to create a unique data visualization.

RQ2: Does attention span level differ by screen time bracket?

- 1.) Group the dataset by screen time bracket and attention span level, then count how many respondents fall into each combination.
- 2.) Using the Plotly library, create a single line plot, with screen time bracket on the x-axis, respondent counts on the y-axis, and a separate line for each attention span level.
- 3.) Compare the lines across brackets to identify differences in attention span distribution as screen time increases.

This will allow for a conclusion on the second research question, by visually analyzing how attention span

levels change based on screen time. By plotting a separate line for each level, it is visually simple to see which lines rise or fall based on how many respondents fall into that category. This also aims for the challenge goal, as it uses an external library to create a unique data visualization.

RQ3: How does screen time differ by education level?

- 1.) Group the dataset by screen time bracket and education level, then count how many respondents fall into each combination.
- 2.) Using the Plotly library, create a single horizontal stacked bar chart, with respondent counts on the x-axis, screen time bracket on the y-axis, and bars colored by education level.
- 3.) Compare the bar segments across brackets to see how screen time distribution differs between education levels.

This will allow for a conclusion on the third research question, by seeing how large each colored segment is for different screen time intervals, for different education levels. That will visually show which education levels contain the most, and the least screen time levels. Similar to the previous two research questions, this also uses an external library to develop a data visualization, following the challenge goal.

RQ4: Are certain age groups more likely to have a higher screen time?

- 1.) Convert each screen time bracket into a single ordinal number, so it can be used as the target the model tries to predict.
- 2.) Convert Age Group into a numeric format (ordinal encoding, since the age groups have a natural order) so the model can use it as an input.
- 3.) Split the data into a training set and a testing set, using Age Group as the input and screen time (the ordinal version) as what the model is trying to predict.
- 4.) Train an Ordinal Logistic Regression model (LogisticAT) on the training set, then use it to predict both the most likely screen time bracket and the probability of each screen time bracket for every age group. This model is used because screen time brackets are ordered categories rather than plain numbers, and ordinal regression respects that ordering better than a standard regression or classifier would.
- 5.) Compare the predicted screen time trend and probabilities across age groups to see if there's a clear pattern, and check how accurate the predictions are using mean absolute error.

This will allow for a conclusion for the fourth research question, by using a machine learning model to predict on data. These predictions will showcase a trend that can be used to highlight if certain age groups are more likely to have a higher screen time, and specifically which ones. This follows the challenge goal as it uses a machine learning model to analyze a specific trend for a conclusion.

### EDA Summary:

For my EDA, I developed a much better understanding of my dataset, and the various patterns that are emerging from it. From my EDA, I also answered 3 out of 4 my research questions by developing visualizations using the Python library Plotly. I analyzed the relationship between screen time, productivity, attention span, and education level. The one main thing that I discovered between all of these visualizations was that there was a non-linear relationship between the variables. There wasn't clear pattern, but rather a mix and a subtle pattern in the data. Despite this sounding insignificant, impactful insight can still be gathered to help answer the essence of this project: How does screen time affect other aspects and factors of life?

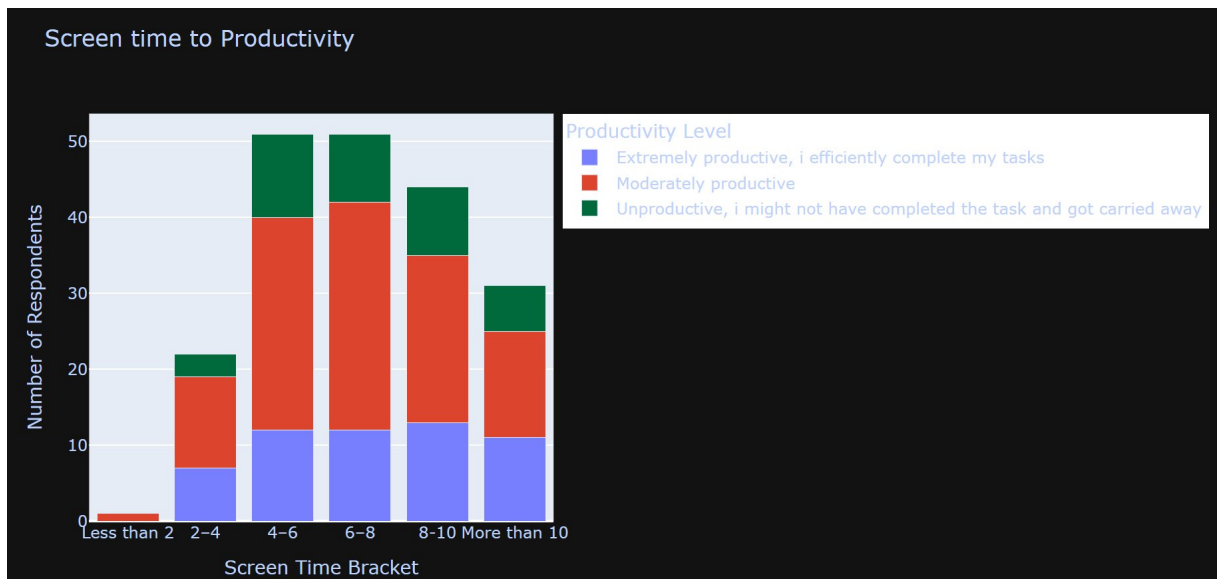
While clear linear relationships were not found through this exploration, it did in fact provide detail into the spread of the data. For instance, I found that the majority of the respondents in the average screen time vs education level visualization was undergraduate students. This type of pattern also occurred across the other two visualizations that I created. Another example is that for my screen time to productivity, I noticed that the majority of the data was moderately productive, being the highest in the majority of the screen time level intervals.

### Results:

#### 1.) Does productivity level differ by screen time bracket?

I discovered a very different result than what I had originally predicted. Specifically, I believed that screen time brackets would be higher when productivity levels were lower. My reason for this was since higher screen time could mean more scrolling, more watching TV, or playing video games, it would link to lower probability. However, I also considered the alternative case of higher screen time linking to higher studying and working hours, thus relating to higher productivity levels.

Despite these predictions my visualization revealed no clear pattern or linear relationship between these two variables:



**Caption:** This is a stacked bar graph that is meant to show how screen time brackets related to productivity levels.

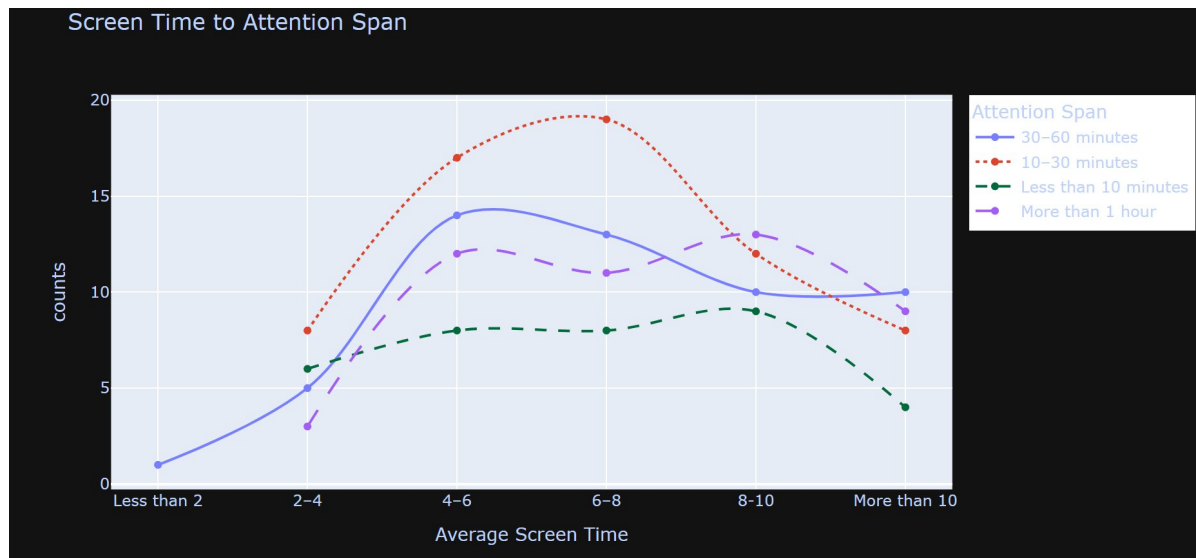
As you can see moderately productive participants are clustered in every single screen time bracket. And the other productivity levels stay relatively consistent. However, there are a couple patterns that can be uncovered from this graph, but it is not extremely clear. For example, we notice that the highest count for unproductive participants is at 4-6 hours and stays relatively the same until 8-10 hours. This shows that the most unproductive participants typically have a higher screen time of 4 hours, going all the way up to 10 hours.

There a couple of reasons why I believe that this visualization came the way it did. First of all, the way productivity was measured might not be what I initially predicted. Specifically, I viewed productivity when compared to doing tasks related to work, or for educational purposes. However, productivity could be anything, whether it's chores, or being productive in a video game. The second reason that I believe my predictions were wrong are because the data might be biased.

The majority of the participants in this dataset are student, meaning they are most likely able to be productive despite their screen time. Alternatively, this could be because these participants utilize study methods online, using screen time.

## 2.) Does attention span level differ by screen time bracket?

For this research question my prediction was somewhat met. Specifically, I predicted that screen time levels would increase, the lower the attention span would be in participants. This can be seen through the lowest attention span (10-30 minutes) rising as the interval of screen time increased:



**Caption:** This graph is created to allow readers to interpret how average screen time fluctuates with attention span, and its main connection.

As you can see, the red line, starts to slowly increase from around 7 counts, to peaking at around 17 counts at 6-8 hours. This trend does indeed support my prediction. However, the line also starts to decrease after 8 hours, clearly contradicting my claim. Along with that, my prediction was also that higher attention span would correlate to lower average screen times. However, when viewing the purple dotted line, (the highest attention span), stays relatively consistent across the majority of screen time levels, and peaks at 8-10 hours of screen time. Overall, this graph provides a contrast and contradiction to what prediction was for the expected value.

One reason why this trend didn't significantly support my original claim, is based on what type of screen time device is most commonly used by participants, and their education level. Specifically, if the most common device is a tablet, and their education level is undergraduate, it is likely that that tablet is being used as a study or note taking tool.

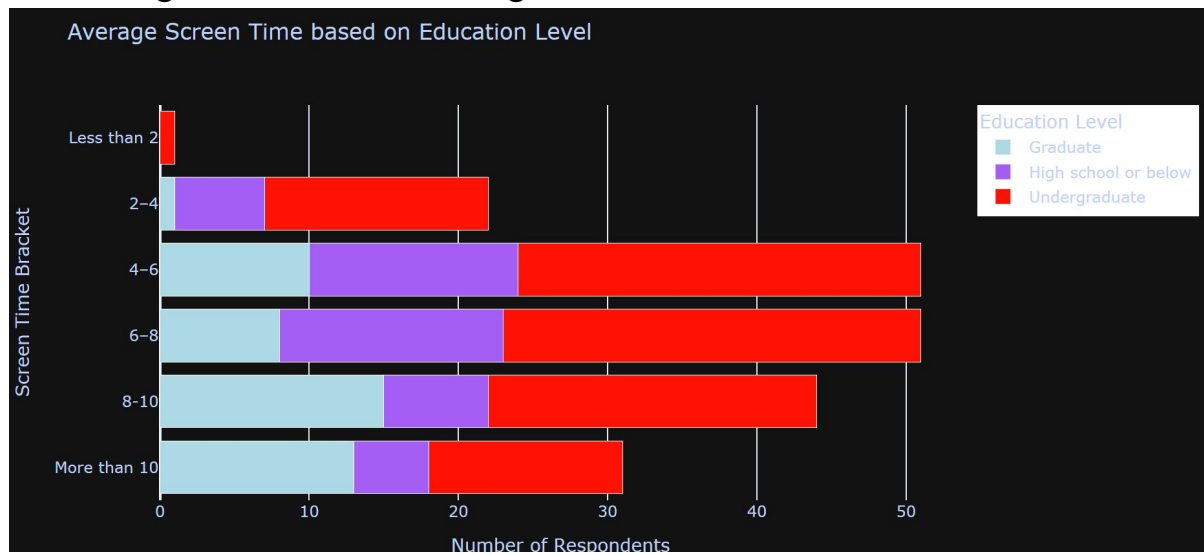
Attention span is most likely to decrease by doing high stimulating, and high dopamine boosting activities. Studying is an activity that is not related to these factors. With this, it would make sense that screen time is high for studying, but attention span is very affected.

This conveys an insightful idea of attention span actually being able to increase or stay constant despite screen time levels. Contradicting my initial prediction, attention span may not be directly linked to screen time levels, but rather the specific activity being done in the screen.

### 3.) How does screen time differ by education levels?

For this research question, my visualization gave me a clear pattern. Specifically, it is

noticeable that undergraduates dominate the screen time brackets from 4-6, all the way up to 8-10. However, as screen time intervals increase, it can be seen that there are higher counts of graduate students for higher screen time intervals:



**Caption:** This horizontal stacked bar graph highlights how education level affects the daily average screen time.

It can also be seen that there is almost no participants in any education level that have a screen time below 2 hours daily. And for participants that are in high school or below, it can be seen that they are clustered mostly in the 4-6 and 6-8 daily hour range. With this, it can be seen that people of higher education levels may tend to have higher daily screen time due to a variety of reasons. First, this could be due to having a higher work load, or it could be based on the current advancements in our current society. Specifically, after going through college, students will likely have a good understanding of digital devices, as they are rapidly advancing in this generation, and are rapidly being used by college students. By understanding the benefits and uses of each technology, screen time might be higher, but be used more effectively and efficiently. For example, one may study using a laptop, and instead reading a physical book, one might recognize the benefits of a kindle, and thus gain more daily screen time.

#### 4.) Are certain age groups more likely to have a higher screen time?

I created a machine learning model to predict which age groups would have which screen time based on the data.



|              | Less than 2 | 2–4   | 4–6   | 6–8   | 8–10  | More than 10 |
|--------------|-------------|-------|-------|-------|-------|--------------|
| Below 18     | 0.9%        | 11.0% | 31.6% | 25.1% | 20.4% | 11.1%        |
| 18–24        | 0.6%        | 8.4%  | 27.1% | 25.5% | 23.9% | 14.5%        |
| 25–34        | 0.5%        | 6.3%  | 22.7% | 24.8% | 27.1% | 18.7%        |
| 35–44        | 0.4%        | 4.8%  | 18.4% | 23.1% | 29.6% | 23.7%        |
| 45 and above | 0.3%        | 3.6%  | 14.7% | 20.7% | 31.1% | 29.6%        |

As you can see, this data frame holds the specific probability of each age group having a daily screen time usage. Again, this contradicts and does not support prediction and beliefs about this research question. This data showcases that the age group 45 and above, have the highest probability of having a screen time hour interval of 8-10, or more than 10.

### *Mean Absolute Error of Model*

#### *1.1*

My MAE is not highly accurate, but the model's predictions were off by about one screen time bracket on average. This is better than chance, suggesting Age Group does carry some predictive signal, but not precise enough to treat individual predictions as highly reliable.

But even if every screen time was off by one bracket, this data shows a pattern.

Specifically, the probability of having a higher screen time correlates with age. As age increases, the higher the probability is for that participant to have a higher screen time. This shows a general trend that can still be used for conclusions.

Based on this trend, there are multiple reasons on why older participants might have higher screen times. Initially, I thought that since younger participants would be more likely to play video games, watch TV, study, or scroll, they would have the higher chance of having a higher screen time. However, as people become older, especially after 45, reaching retirement age, can lead to significant free time. This time can be allotted to watching TV or doing other fun activities.

This claim also works the other way around, that higher aged individuals are more likely to have more work from jobs or daily tasks, meaning that in this time period, they are extremely likely to use all sorts of digital technologies across the day. Take a software engineer as an example, they have an average of 10-14 hours of screen time, factoring in both personal and professional use. This compared to a child who might be playing sports, or spending their time at school where computers might not be used that often.



Average predicted screen time by age group:

|              |   |
|--------------|---|
| Below 18     | 3 |
| 18-24        | 3 |
| 25-34        | 3 |
| 35-44        | 4 |
| 45 and above | 4 |

dtype: int64

"Less than 2": 0,

"2-4": 1,

"4-6": 2,

"6-8": 3,

"8-10": 4,

"More than 10": 5

This result, showing the specific model predictions, also shows that higher aged individuals tend to have higher average daily screen time usages.

### Impact and Limitations:

To begin, my results benefit groups in a variety of ways. Specifically, showing that productivity levels and attention span may not be directly related to the amount of screen time, can help multiple people attempting to fix their lives. For instance, many students struggle with phone addiction, or an excessive amount of phone usage. One might think that it is just dependent on the number of hours on the screen, however, after these results, they can now account for not the hours, but the type of activity being done on the phone. Is it reading? Is it studying? Or is it watching Instagram Reels? This is a more significant factor when it comes to determine whether one has higher or lower productivity levels or attention span.

In addition, my results should be used as a general trend, or pattern in participants, not for specific individuals. For my model, there was a significant error, which means that while the general direction of how screen time increases based on age might be correct, the individual and specific hours might be misleading. This means that this will harm people looking for an understanding of a specific screen time bracket. One might be attempting to understand the real difference between 4-6 hours of screen time and 8-10 hours of screen time. But all of my results suggest a very subtle difference, and not a clear, distinct pattern.

One group of people that might be excluded from this are very young kids. The majority of the participants in the dataset that I am using are 18 and older. In fact, only about 17% of participants are under 18. This creates a bias towards older participants and doesn't provide realistic and accurate trends or predictions for young kids. My results won't significantly benefit people who are attempting to analyze the affects on their young toddlers, for example.

Finally, one crucial takeaway from my results, are that there is not clear, linear pattern or relationship between most of the two variables that I analyzed. This means that the patterns are very subtle, or not very common or significant to a point where they can't really be generalized to a large public.

## Challenge Goals:

Selected 2 Goals:

### 1.) Incorporating a new Python Library

I believe that this project will meet this goal the research questions developed for this project will require data visualization for effective analysis. By importing new types of data visualization libraries, this project will be to use more unique types of data visualizations to further enhance the effectiveness of the message conveyed. This project will meet this goal because it will further illustrate specific patterns about how screen time affects a variety of factors.

### 2.) Using Machine Learning

Part of this project is to analyze the specific decisions or preferences made based on both the details of a respondent, but also their screen time levels. With these variables, incorporating machine learning to predict possible scenarios using models will be effective on testing on how reliable, or how strong a trend is in this dataset. This project will meet this goal because it will further emphasize a specific trend, or pattern seen in this data.

## Testing:

I tested my research questions by having one test module creating a smaller dataset mimicking the original dataset, and checked if my visualizations were correct. By using a smaller dataset I checked whether the general pattern of how my data was distributed in the visualizations using the larger dataset was matching up. I then used multiple assert statements and methods to check whether the mappings on my model were correct, if all classes were learned from my model, and checked whether the probability percentages added up to 100. These were in 2 separate files.

## Work Plan Evaluation:

My work plan worked exceptionally well, but it took a little less time than I had originally planned. Plus, due to some issues during code, I was forced to change some of the methodology and visualizations that I was creating. For example, instead of creating multiple pie charts, that would be extremely inconvenient, I created a simple bar graph that evaluated everything at once that was much more efficient. This did take a little more research and planning, however once this one step was implemented, the rest of the steps were much easier and faster to do, making my each step estimated time an overestimate. In addition to that, I did change the specific machine learning model that I used, and used more specific types of methods in libraries that we already learned, to create a more enhanced version of my project. Other than that, my timings for each task was relatively equal to what I had predicted.